

Fusion, Unified Embeddings, and Data Scarcity

DATA 443/643 · Week 6, note 6.3

Week 6

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Ideas from CMU 11-777 (early / late / cross-attention fusion) and note 4.1's coordinated baseline. Original slides; that course is not copied.

Lecture plan

- 1 Early, late, cross-attention
- 2 Concat grows width; add needs a shared axis
- 3 The data pyramid
- 4 What to write in a report

Key goal: name the fusion and the missing-modality plan. Cosine is not cross-attention.

First time: what / why / formula

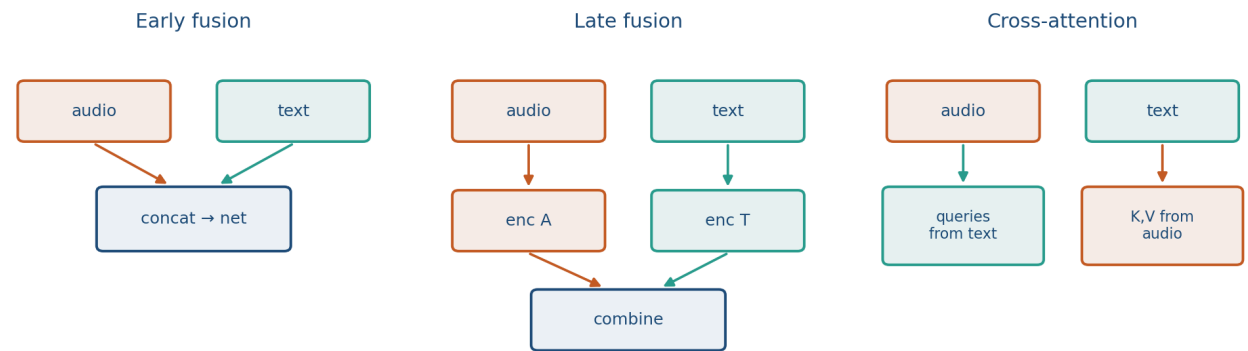
- What / why. Early fusion (concat features then one net), late (combine decisions), cross-attention (queries from one side, keys/values from the other). Concat is the student default and grows width. Missing a modality at test time breaks naive concat.
- Architecture. Draw two towers, then pick concat / add / cross-attn. Data pyramid: lots of unpaired text, less paired A/V.
- Formula. Concat: $[u;v] \in \mathbb{R}^{d_u + d_v}$. Add needs $d_u = d_v$. Cross-attn: $\text{softmax}(Q_u K_v^\top / \sqrt{d}) V_v$.
- Tradeoffs. + Cross-attn can attend to a spectrogram while generating text. – Width, missing-modality failure, and paired-data cost.

How. Lab 6: concat vs add on toy vectors. Project: name the fuse and what you do if audio is missing.

Three places to fuse

- Early: concat features, one backbone. Brittle if audio is missing.
- Late: a model each, then combine scores.
- Cross-attention: queries from one stream, K and V from the other.

Three ways to mix streams. Missing audio at test time punishes early fusion.



Concat versus add

- $[a; v] \in \mathbb{R}^{256}$ if each is 128-D.
- Add is illegal if widths differ. Map first: $W_v \in \mathbb{R}^{128 \times 256}$.
- CLIP two-towers never concat pixels with token ids.

If widths differ, add is illegal until you map. Missing audio breaks concat unless you trained a mask.



A missing microphone

$[a;v] \in \mathbb{R}^{256}$ needs both streams

Early concat with no dropout assumes audio is there.

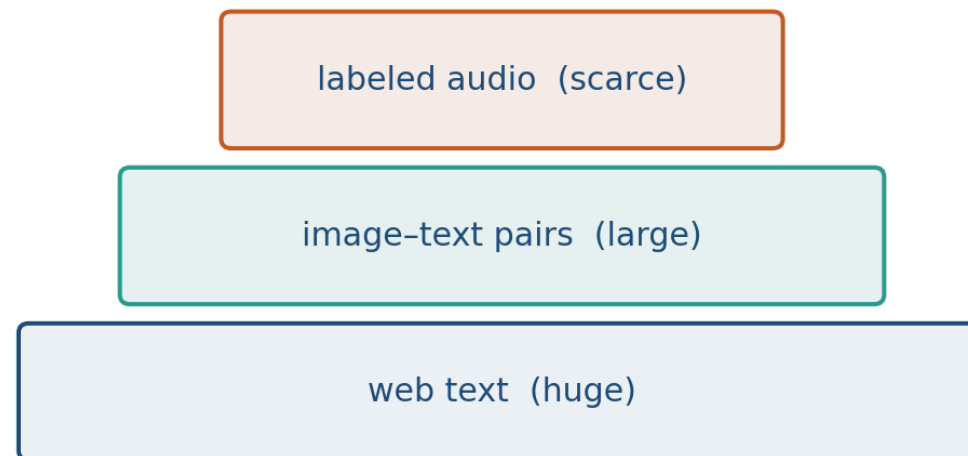
Late towers can skip a cosine if the mic dies.

Cross-attention can mask a whole key set.

Pretrain where the data is thick

- Web text, then image–text, then transcribed audio, then labeled tasks.
- Freeze Whisper or CLIP. Train a small head on 400 rows.
- Do not train a joint stack from scratch on 400 clips.

Paired, transcribed audio is the thin top of the data pyramid



What to take from this lecture

Name it

Early, late, or cross-attention.
Which loss sees both streams?

Count it

Unpaired hours, paired hours,
labeled rows.

Lab 6

Concat versus add in 4-D is this
slide in miniature.